Annexure 03

General Sir John Kotelawala Defence University
Faculty of Computing
IT42999 - Individual Research Project

APPLICATION FOR UNDERGRADUATE FINAL YEAR RESEARCH PROJECT

1. Project Title *The title of the project should be brief, reflect concisely and accurately the proposed. Students are advised to avoid titles which convey a distant or potential application of the proposed work, or a greater aspiration or goal than is to be expected from the proposed work.*	
Design and Evaluation of an Agentic AI-Driven Decision Support System for Explainable Early Disease Risk Detection in Companion Animals	
2. Research Area	**Specific Area**
Artificial Intelligence Information Technology	Agentic AI and Intelligent Decision Support Systems Machine Learning for Animal Health Monitoring Mobile Application Development for Veterinary Support
3. Supervisor(s)	
Name and Designation: Dr. (Mrs) N Wedasinghe Senior Lecturer (Faculty of Computing, Department of Information Technology) Institution (if not from the University): Area of expertise related to the proposed project:	Mobile : +94722601379 E-mail :nirosha@kdu.ac.lk
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4. STUDENT	
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5. Summary

(a) Explain briefly the research problem, research approach and expected outputs.
Not exceeding one A4 page, font type Times New Roman, size 11, single space.

Companion animals such as dogs and cats are vulnerable to various chronic and life-threatening diseases that often develop gradually with subtle or non-specific symptoms. Since animals cannot verbally express discomfort, early warning signs such as changes in appetite, water intake, activity level, or urination patterns are frequently overlooked by pet owners. As a result, veterinary care is often sought only after the disease has progressed to an advanced stage, reducing treatment effectiveness and increasing healthcare costs.

This project proposes an Agentic AI-driven mobile decision support system that analyzes observable symptoms and behavioral patterns of companion animals to identify early disease risk. The system employs machine learning techniques for risk prediction and an agentic AI layer to autonomously interpret results, assess urgency, and recommend suitable actions. These actions include guiding pet owners to nearby veterinary clinics or hospitals and displaying available veterinary surgeons with time slots. The solution is designed as a decision support tool rather than a diagnostic system, aiming to promote early awareness, timely veterinary consultation, and improved animal welfare.

(b) Give 3 – 5 keywords for the proposed project:

Agentic AI, Decision Support System, Machine Learning, Pet Health Monitoring, Early Disease Risk Detection

6. Research Problem

6.1 Research problem/s

Companion animals often suffer from life-threatening diseases that show limited or non-specific symptoms in their early stages. Pet owners lack accessible tools to identify early disease risk based on observable behavioral and physical changes, resulting in delayed veterinary consultation and advanced disease progression.

6.2 Analysis of the problem/s & rationale for the research question

Early disease detection in companion animals remains a significant challenge due to the indirect nature of symptom expression and reliance on behavioral and physiological indicators. Subtle changes in appetite, activity, hydration, and elimination patterns are often overlooked or misinterpreted by pet owners, leading to delayed veterinary consultation and advanced disease progression. Existing digital pet health tools primarily offer static information or manual symptom checklists, lacking autonomous reasoning, real-time analysis, and actionable decision support.

Moreover, current systems do not integrate preventive care functions such as vaccination tracking or real-time access to veterinary services, limiting their effectiveness in promoting timely intervention. The absence of structured, intelligent risk assessment mechanisms reduces owner confidence and weakens the adoption of preventive veterinary practices. These challenges are particularly pronounced in contexts where access to veterinary expertise is limited due to geographical, financial, or infrastructural constraints.

This research is therefore motivated by the need to develop an accessible, intelligent, and proactive decision support system that leverages Agentic Artificial Intelligence to autonomously analyze symptom and behavioral data, assess early disease risk, and guide timely veterinary action. By integrating risk prediction, preventive care reminders, and real-time veterinary service recommendations within a mobile platform, the proposed system aims to bridge the gap between symptom observation and effective clinical response, thereby improving animal welfare outcomes and promoting responsible pet ownership.

7. Comprehensive literature review AND the complete list of references in the relevant area.
(Attach additional sheets if necessary)

(Kim & Kim, 2024) developed an AI-based health score system that uses sensor-collected behavioral data to quantitatively assess a dog's health status. The Health Score correlated highly with veterinary assessments (≈87.5% agreement), demonstrating a data-driven, pet-owner-accessible health monitoring model. Unlike static clinical models, the proposed project extends this by integrating Agentic AI reasoning to interpret health risk patterns and trigger personalized alerts for disease risk and vaccination timing. [1]

(Szlosek et al., 2024) validated a machine learning algorithm to distinguish preventive wellness visits from other veterinary visits using clinical data. The model showed high specificity (0.94) and sensitivity (0.86), indicating its potential in subclinical disease identification during wellness assessments. The proposed system adapts this concept to real-world pet symptom patterns, applying mobile-collected data and Agentic AI reasoning to detect early disease risks before clinical visits [2]

(Das et al., 2024) reviews the application of various machine learning techniques in animal health, highlighting gaps in early detection methods and the need for real-time, accessible models. The survey finds that ML can improve early disease detection with appropriate datasets, which aligns with the need in your project to leverage owner-reported symptoms and machine learning for early risk identification [3]

(Jokar et al., 2024) explores the potential and challenges of using AI chatbots to assist pet owners with health information and decision-making. While not a full decision support system, it highlights how conversational AI can provide early guidance based on symptoms, supporting the notion that AI can enhance pet owner decision processes a concept extended in your Agentic AI system [4]

(Aguilar-Lazcano et al., 2023) examines sensor fusion and machine learning for understanding animal behavior. Although focused on multi-species monitoring and welfare, it demonstrates how behavioral data can be processed for animal health insight a key component underlying symptom and behavior analysis in your project's predictive models [5]

(Imada et al., 2024) reports the application of machine learning in predicting multiple disease types in cats and dogs, including early risk patterns and survival rates. It notes that models often serve as predictive algorithms rather than standalone diagnostics, similar to your project's goal to use ML for risk stratification integrated with Agentic decision support [6]

(Eman et al., 2025) highlights molecular biomarkers for early disease detection, pointing out the challenge of late-appearing clinical signs. Although biomarkers are lab-centric, the concept emphasizes the need for early signals, reinforcing the importance of symptom and behavioral data when lab data are unavailable as in your mobile system [7]

This article from the Journal of Computer Science, (Rathi & Sumathi, 2025) reviews how machine learning assists in classifying various pet diseases (e.g., diabetes, arthritis, kidney disease) using datasets. It underscores the promise of ML for early detection and risk estimation, supporting your project's ML risk models[8]

(Bouchemla et al., 2023) review identifies AI use in diagnostic and decision-support roles in veterinary medicine. It highlights AI expert systems in hematology, imaging, and algorithmic support, supporting the broader context for your Agentic AI system's decision support layer.[9]

Aishwarya et al. (2025) proposed PAWPAL, an AI-based pet health assistance system that combines image-based disease detection using CNNs with NLP-based symptom analysis to provide early guidance to pet owners. The system achieved high accuracy in identifying common skin diseases and offered real-time care recommendations. However, PAWPAL primarily focuses on image

classification and symptom text input, whereas the proposed project extends this work by incorporating Agentic AI, behavioral pattern analysis, autonomous decision support, vaccination tracking, and real-time veterinary service discovery for broader and proactive health risk management.[10], [11]

This open access paper, (Jin et al., n.d.) examines how AI tools automatically extract symptoms from clinical records and predict disease, demonstrating ML's role in veterinary decision support and workload reduction. The proposed project builds on this by using owner-entered data instead of clinical records and adding Agentic analytics.[12]

Shinde et al. (2025) reviewed how AI is used to predict and detect animal diseases using images, sensors, and behavioral data. The study shows that AI can help identify diseases early and improve animal health management. It also discusses challenges such as data quality and system reliability. Compared to this review-based study, the proposed project applies these AI ideas practically by developing an Agentic AI mobile system focused on early disease risk detection in companion animals with real-time veterinary support and vaccination tracking.[13], [14]

Kulkarni et al. (2025) proposed a machine learning-based framework for predicting animal diseases using environmental, behavioral, genetic, and historical health data. The system supports real-time monitoring and proactive disease management, mainly focusing on livestock health and outbreak prevention. It improves veterinary decision-making and reduces economic losses in agriculture. Compared to this study, the proposed project focuses specifically on companion animals and implements an Agentic AI-driven mobile application for early disease risk detection, vaccination tracking, and real-time veterinary service guidance.[15]

(Akinsulie et al., 2024) discusses how AI supports predictive epidemiology, precision diagnosis, and early intervention, highlighting AI's capability to manage complex animal health data. This research supports the theoretical foundation for using AI to predict risk before clinical manifestation in pets[16]

References

[1] S. C. Kim and S. Kim, "Development of a Dog Health Score Using an Artificial Intelligence Disease Prediction Algorithm Based on Multifaceted Data," *Animals*, vol. 14, no. 2, Jan. 2024, doi: 10.3390/ani14020256.

[2] D. Szlosek, M. Coyne, J. Riggott, K. Knight, D. J. McCrann, and D. Kincaid, "Development and validation of a machine learning model for clinical wellness visit classification in cats and dogs," *Front. Vet. Sci.*, vol. 11, 2024, doi: 10.3389/fvets.2024.1348162.

[3] S. Das, R. K. Roy, and T. Bezboruah, "Machine Learning in Animal Healthcare: A Comprehensive Review," *International Journal of Recent Engineering Science*, vol. 11, no. 3, pp. 89–93, Jun. 2024, doi: 10.14445/23497157/ijres-v11i3p109.

[4] M. Jokar, A. Abdous, and V. Rahmanian, "AI chatbots in pet health care: Opportunities and challenges for owners," May 01, 2024, *John Wiley and Sons Inc*. doi: 10.1002/vms3.1464.

[5] C. A. Aguilar-Lazcano, I. E. Espinosa-Curiel, J. A. Ríos-Martínez, F. A. Madera-Ramírez, and H. Pérez-Espinosa, "Machine Learning-Based Sensor Data Fusion for Animal Monitoring: Scoping Review," Jun. 01, 2023, *MDPI*. doi: 10.3390/s23125732.

[6] J. Imada, J. C. Arango-Sabogal, C. Bauman, S. Roche, and D. Kelton, "Comparison of Machine Learning Tree-Based Algorithms to Predict Future Paratuberculosis ELISA Results Using Repeat Milk Tests," *Animals*, vol. 14, no. 7, Apr. 2024, doi: 10.3390/ani14071113.

[7] S. Eman *et al.*, "Technologies in Biomarker Discovery for Animal Diseases: Mechanisms, Classification, and Diagnostic Applications," Nov. 01, 2025, *Multidisciplinary Digital Publishing Institute (MDPI)*. doi: 10.3390/ani15213132.

[8] J. Rathi and A. Sumathi, "Animal Health Prediction Using Hybrid KNN Based Vector Neighbor Classification Model: A Machine Learning Approach," *Journal of Computer Science*, vol. 21, no. 9, pp. 2088–2095, Sep. 2025, doi: 10.3844/jcssp.2025.2088.2095.

[9] F. Bouchemla, S. V. Akchurin, I. V. Akchurina, G. P. Dyulger, E. S. Latynina, and A. V. Grecheneva, "Artificial intelligence feasibility in veterinary medicine: A systematic review," *Vet. World*, vol. 16, no. 10, pp. 2143–2149, 2023, doi: 10.14202/vetworld.2023.2143-2149.

[10] A. AP, "PAWPAL: AI-Based Pet Health Assistance System Using Image-based Disease Detection and NLP Symptom Analysis," *Int. J. Res. Appl. Sci. Eng. Technol.*, vol. 13, no. 12, pp. 2065–2072, Dec. 2025, doi: 10.22214/ijraset.2025.76461.

[11] A. AP, "PAWPAL: AI-Based Pet Health Assistance System Using Image-based Disease Detection and NLP Symptom Analysis," *Int. J. Res. Appl. Sci. Eng. Technol.*, vol. 13, no. 12, pp. 2065–2072, Dec. 2025, doi: 10.22214/ijraset.2025.76461.

[12] Y.-G. Jin *et al.*, "Date of publication xxxx 00, 0000, date of current version xxxx 00, 0000. AI Veterinary Assistance: Enhancing Clinical Decision-making in Animal Healthcare", doi: 10.1109/ACCESS.2024.0429000.

[13] O. Shinde, S. Sankpal, Y. Kandalkar, T. Bhosale, and S. Khairnar, "Emerging role of artificial intelligence in animal disease surveillance, prediction and diagnosis: A Review," *International Journal of Veterinary Sciences and Animal Husbandry*, vol. 10, no. 10S, pp. 41–45, Jan. 2025, doi: 10.22271/veterinary.2025.v10.i10Sa.2623.

[14] A. I. Pereira *et al.*, "Artificial Intelligence in Veterinary Imaging: An Overview," May 01, 2023, *MDPI*. doi: 10.3390/vetsci10050320.

[15] P. P. P. Kulkarni, K. Jay, K. Pratik, and I. Kshitij, "Animal Disease Prediction Using Machine Learning," 2025.

[16] O. C. Akinsulie *et al.*, "The potential application of artificial intelligence in veterinary clinical practice and biomedical research," *Front. Vet. Sci.*, vol. 11, 2024, doi: 10.3389/fvets.2024.1347550.

Comparison of Existing Pet Health Applications vs Proposed System

Feature/ Aspect	Existing Pet Health Applications	Proposed System
Symptom Handling	Manual symptom checklists; limited behavioral analysis	Structured symptom, behavioral pattern analysis with intelligent interpretation
Disease Risk Prediction	Mostly descriptive info; no prediction	Machine learning–based early disease risk prediction
Use of AI	Basic rule-based or static logic	Agentic AI with autonomous reasoning, prioritization, and action triggers
Preventive Care	Reminders only	Intelligent preventive care agent (vaccination tracking, risk-aware reminders)
Decision Support	Provides general information only	Action-oriented decision support (risk alerts, recommendations, urgency assessment)
Behavior Monitoring	Usually missing or very limited	Multi-factor behavior, symptom pattern analysis for risk assessment
User Guidance	Generic advice	Personalized guidance based on risk level, pet profile, and detected patterns
System Intelligence	Static; no autonomous action	Agentic layer that generates alerts and recommendations without user prompts
Data Sources	Single-type inputs	Hybrid input: symptoms, behaviors, demographics, preventive health data
Veterinary Support	Manual clinic search or text lists	Automated location-based vet discovery, surgeon availability, real-time time slots

8. Originality & innovativeness of the proposed work

The proposed work is original and innovative as it focuses on early disease risk detection in companion animals rather than clinical diagnosis. Unlike many existing pet health systems that depend on laboratory tests, medical imaging, or veterinary input, this project utilizes owner-observed symptoms and behavioral patterns, making the solution more accessible and practical for everyday use.

The system introduces Agentic AI to autonomously analyze symptom data, assess disease risk, and guide users toward appropriate actions such as seeking veterinary care. Additionally, the integration of disease risk awareness with real-time discovery of nearby veterinary clinics and available surgeons enhances the practical value of the application. The emphasis on preventive care, decision support, and early awareness distinguishes this project from traditional pet health applications and contributes to improved animal welfare.

Key innovative aspects include:

- Application of Agentic AI for autonomous and intelligent veterinary decision support
- Integration of early disease risk detection with real-time veterinary clinic and surgeon availability
- Focus on preventive care and early awareness rather than medical diagnosis

9. Overall aim and specific objectives of the proposed work

9.1 Overall aim

To develop an Agentic AI-driven decision support system for early disease risk detection in companion animals.

9.2 Specific Objective/s

- To identify early disease risk in companion animals using owner-observed symptoms and behavioral patterns
- To develop and evaluate machine learning–based models for disease risk prediction
- To implement an Agentic AI framework that autonomously reasons over risk levels and provides action guidance
- To integrate location-based veterinary clinic and surgeon availability with real-time time-slot information
- To design a vaccination management and reminder mechanism to support timely preventive care for companion animals

10. Methodology

10.1 Describe the Methodology
(*Attach additional sheets if necessary)*

This study follows a Design Science Research approach, as the primary objective is to design, develop, and evaluate an Agentic AI-driven decision support system for early disease risk detection in companion animals. A mixed-method research approach is adopted, combining quantitative and qualitative techniques. Quantitative methods are used to train and evaluate machine learning models using structured symptom and behavioral data, while qualitative feedback is used to assess system usability, trust, and practical effectiveness. The system development follows an iterative SDLC model, allowing continuous refinement based on testing and user feedback.

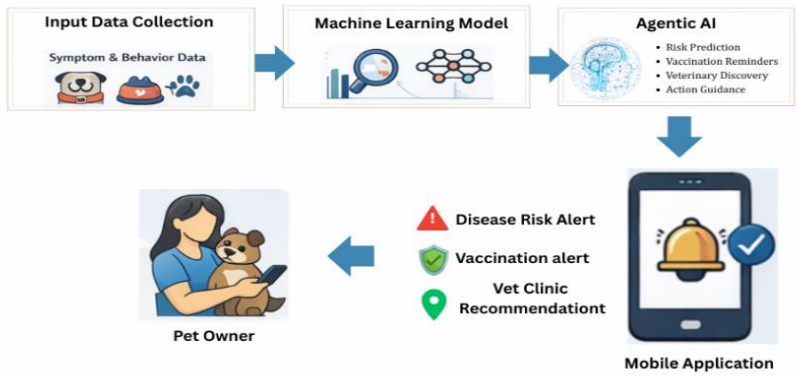
Primary data is collected through structured online surveys from dog and cat owners, capturing observable symptoms such as appetite, water intake, urination frequency, activity level, vomiting, vaccination status, and basic pet demographics. Secondary data from validated literature sources and public datasets are used to enhance dataset quality and balance. The collected data is pre-processed, encoded, and used to train machine learning classification models. An Agentic AI layer interprets model outputs, assesses urgency, and autonomously generates actions such as disease risk alerts, vaccination reminders, and recommendations for nearby veterinary clinics and available surgeons.

The system is implemented as a mobile application supported by a backend server. Model performance is evaluated using accuracy and validation metrics, while system effectiveness is assessed through functional and usability testing. Ethical standards are maintained through voluntary participation, anonymized data handling, and the exclusion of sensitive personal information.

10.2 Experimental design where applicable
Please complete relevant sections

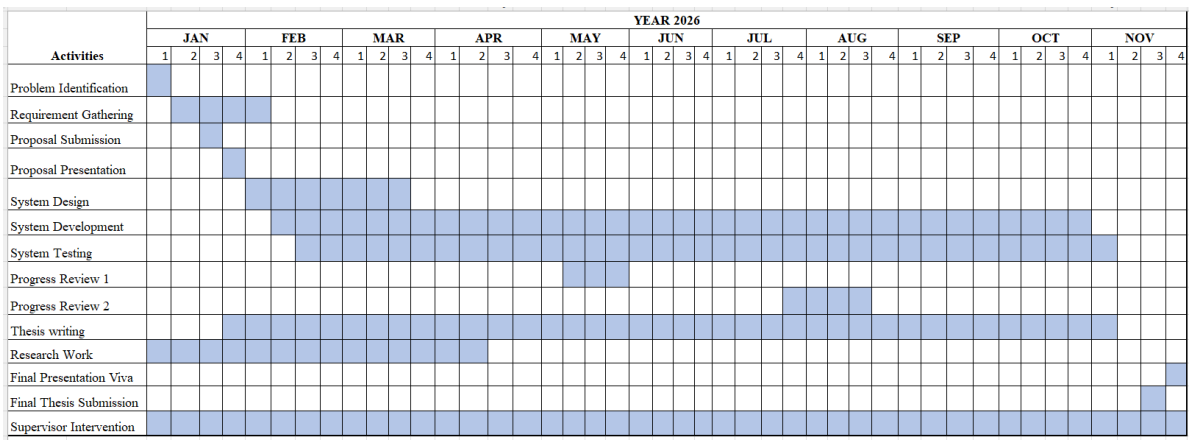
The system begins with collecting symptom and behavioral data from pet owners through the mobile application. This data is processed using machine learning models to predict early disease risk. The Agentic AI layer then reasons over the prediction results to generate appropriate actions such as risk alerts, vaccination reminders, and recommendations for nearby veterinary clinics and available

surgeons. These outputs are delivered to the pet owner through the mobile application to support timely veterinary decision-making.



10.3 **Work plan**

Please attach the quarterly Gantt Chart to cover the proposed study, as per the format below.



10.4 **Indicators & milestones of progress**

Please list the milestones and indicators that will be used to measure the progress of the proposed Study

1. Completion of research and requirements gathering
2. Approval of the prototype design
3. Successful integration of external tools and technologies
4. Completion of front-end and back-end development
5. Successful user testing and quality assurance
6. Deployment and launch of the platform.
7. Ongoing platform performance and user feedback
8. Achieving desired user acquisition and engagement metrics
9. Completion of user training and documentation
10. Continuously improving the platform based on data analysis and user feedback.

11. Ethical considerations	
Relevance to the project	Relevant ☐ Not relevant ☑

12. Plan to protect the human & environmental safety issues related to the project

- User privacy and safety will be ensured by implementing secure authentication mechanisms and encrypted storage for personal and pet-related data.
- The system will function strictly as a decision support tool and will clearly state that it does not replace professional veterinary diagnosis, reducing the risk of misuse or misinterpretation.
- Clear in-app guidance and alerts will be provided to encourage timely veterinary consultation when high disease risk is detected.
- The mobile application will be designed with a simple and accessible interface to prevent user confusion or incorrect data entry.
- Lightweight machine learning models will be used to minimize device resource consumption and reduce energy usage.
- Compatibility with a wide range of mobile devices will be maintained to avoid frequent hardware upgrades and minimize electronic waste.
- Regular system updates and testing will be conducted to address potential security vulnerabilities and ensure reliable system performance.

13. Expected Research Outputs

Deliverables at the end of the project in point form

- A fully functional mobile-based Agentic AI decision support system for companion animal health monitoring.
- An Agentic AI-driven symptom and behavior analysis module for early disease risk detection.
- A machine learning classification model to identify early risk of selected high-impact diseases.
- An autonomous health monitoring agent that continuously evaluates pet health data and risk patterns.
- A preventive care agent that tracks vaccination timelines and generates intelligent alerts without manual user input.
- A vaccination reminder system integrated with disease risk analysis to support preventive healthcare.
- A real-time veterinary clinic and hospital locator based on user location.
- Display of available veterinary surgeons and consultation time slots.
- A risk-based alert and guidance system recommending timely veterinary consultation.
- Secure user and pet profile management.
- A research report documenting methodology, experimental results, and evaluation.
- Performance evaluation and usability testing results.
- System documentation and user guidelines.

14. Plan to utilize the research output

- Use the system as a decision support and preventive care tool for pet owners.
- Assist pet owners in early disease awareness and vaccination compliance.
- Demonstrate the system to veterinary clinics and animal welfare organizations.
- Promote responsible pet ownership through intelligent health monitoring.
- Publish research findings in academic conferences and journals.
- Extend the system for future features such as wearable data integration.
- Use feedback to enhance Agentic AI reasoning and system accuracy.

15. Research Outcomes

15.1 Significance of research outcomes and the impact on national/socio-economic development of Sri Lanka.
Please write in point form.

• Enhances early disease risk detection in companion animals, supporting timely veterinary intervention and preventive care.
• Reduces disease severity and treatment costs by promoting early awareness and proactive health management.
• Improves animal welfare and longevity, contributing to responsible pet ownership practices in Sri Lanka.
• Demonstrates the practical application of Agentic AI in veterinary decision support systems.
• Strengthens national capacity in applied artificial intelligence and digital health innovation.
• Supports the development of accessible, low-cost digital health tools suitable for local socio-economic contexts.

15.2 Identify relevant stakeholders and potential beneficiaries

- Pet owners
- Veterinary clinics and surgeons
- Animal welfare organizations
- Researchers and students
- Academic institutions

16. Plan to protect and exploit Intellectual Property (IP)?
(Indicate if applicable)

Not Applicable

17. Signature of Student and Supervisor(s)

The details given above are correct and personally verified.

(a) …… (Student)	…18/01/2026…….. Date
(b) …… (Principal Supervisor)	18/01/2026 ……..... Date
(c) …… (Co- Supervisor)	18/01/2026 ……..... Date

18. Recommendation and Approval	
The application is recommended and approved.	
……...……	……..
Department Project Coordinator	Date
……...……	……..
Head of the Department	Date

Guidelines for Applicants

1. The proposed research project should necessarily be an original investigation.
2. The problem to be tackled should be clearly identified.
3. A comprehensive literature review should be done, and all details should be provided together with the list of references.
4. The budget should be calculated correctly and stated in detail.
5. A detailed time-based work plan (Gantt chart) should be included stating the proposed activities and time frames clearly.
6. In the case of selection of a supervisor from an outside institution, it is mandatory to get an internal supervisor appointed.
7. Softcopy of the application should be submitted as PDF format.
8. Incomplete applications **will not** be considered.